Error Handling Helpline

Group Members

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GitHub:

https://github.com/WILR96/ErrorHandlingHelpline

Task board:

https://github.com/users/WILR96/projects/1

Project Idea Summary

This project is a debugging and programming help platform that enables students to share technical problems and receive support from the community. Users can post error messages, code snippets, and environment-related issues, while others can respond with explanations and debugging guidance.

The platform aims to encourage collaborative learning, peer support, and the development of problem-solving and critical thinking skills.

Core User Stories

Create Help Post

As a user, I want to create a help request, so that I can receive assistance with a coding problem.

Acceptance Criteria:

- User must be logged in.
- User can add title and description.
- Optional code/image upload.
- Post appears in main feed with any files added.

View Posts

As a user, I want to browse help posts, so that I can find problems I can assist with.

Acceptance Criteria:

- All posts are displayed in the main feed.
- Each post shows title, author and timestamp.
- Posts are ordered by most recent by default.

Answer a Post

As a user, I want to respond to a help request, so that I can assist another student.

Acceptance Criteria:

- User must be logged in.
- User can submit a written response.
- Response appears under the original post.

- Response shows author and timestamp.

Mark Answer as Solved

As the original poster, I want to mark an answer as solved, so that helpful responses are recognised.

Acceptance Criteria:

- Only the original poster can mark an answer as solved.
- Only one answer can be marked as solved.
- Solved answer is visually highlighted.
- Reputation points are awarded to the responder.

Mark Answer as Helpful

As a user, I want to mark an answer as helpful, so that quality contributions are rewarded.

Acceptance Criteria:

- User must be logged in.
- A user can only mark an answer helpful once.
- Helpful count updates immediately.
- Reputation points are awarded to the responder.

Gamification User Stories

Earn Reputation

As a contributor, I want to earn reputation points when my answers are marked as solved or helpful, so that my contributions are recognised.

Acceptance Criteria:

- Reputation increases when answer is marked solved.
- Reputation increases when answer is marked helpful.
- Reputation cannot exceed daily cap.
- Updated reputation is reflected immediately on profile.

View Reputation

As a user, I want to see my reputation score and role title, so that I can understand my standing within the community.

Acceptance Criteria:

- Reputation score is visible on user profile
- Role title is displayed based on reputation
- Role updates automatically when threshold is reached

Automatic Role Progression

As a contributor, I want my role (e.g., Contributor, Trusted Helper, Community Mentor) to update automatically based on my reputation, so that mentorship emerges naturally.

Acceptance Criteria:

- The system assigns a role based on predefined reputation thresholds
- Role updates automatically when a user reaches a new threshold

- Role changes do not require manual approval
- The updated role is displayed immediately on the user's profile
- The updated role is displayed next to the user's username in posts
- Role cannot be manually edited by the user

View Other Users' Reputation

As a user, I want to see reputation and role titles next to usernames in posts, so that I can identify experienced contributors.

Acceptance Criteria:

- Role title is displayed next to username
- Reputation score is displayed next to username
- Display updates automatically when reputation changes

Earn Badges

As a user, I want to earn badges for milestones (e.g., first solved answer), so that positive behaviour is encouraged.

Acceptance Criteria:

- Badge is awarded when milestone is reached
- Badge is visible on user profile
- Badge remains permanently once earned

Private Messaging

As a user, I want to be able to message other users privately for help, so that I can receive more detailed guidance outside of a public thread.

Acceptance Criteria:

- User must be logged in to send or receive messages.
- User can send a private message to another registered user.
- Recipient receives a notification of a new message.
- Messages are only visible to the sender and recipient.
- Message history is stored and viewable within a conversation thread.
- Users cannot message themselves.
- Users can report or block inappropriate messages.

Moderation User Stories

Report Content

As a user, I want to report inappropriate or low-quality content, so that the community remains supportive and safe.

Acceptance Criteria:

- Report button is visible on posts and answers
- User must be logged in to report
- Report is recorded in the system
- Administrator can review reported content

Daily Reputation Cap

As a system administrator, I want to limit daily reputation gain, so that users cannot exploit the scoring system.

Acceptance Criteria:

- The system defines a maximum amount of reputation that can be earned per user per 24-hour period
- Reputation points stop increasing once the daily cap is reached
- The cap resets automatically after 24 hours
- Users cannot bypass the cap through multiple posts or repeated interactions
- The system still records helpful or solved actions even if no additional reputation is awarded
- The cap value can be configured by an administrator

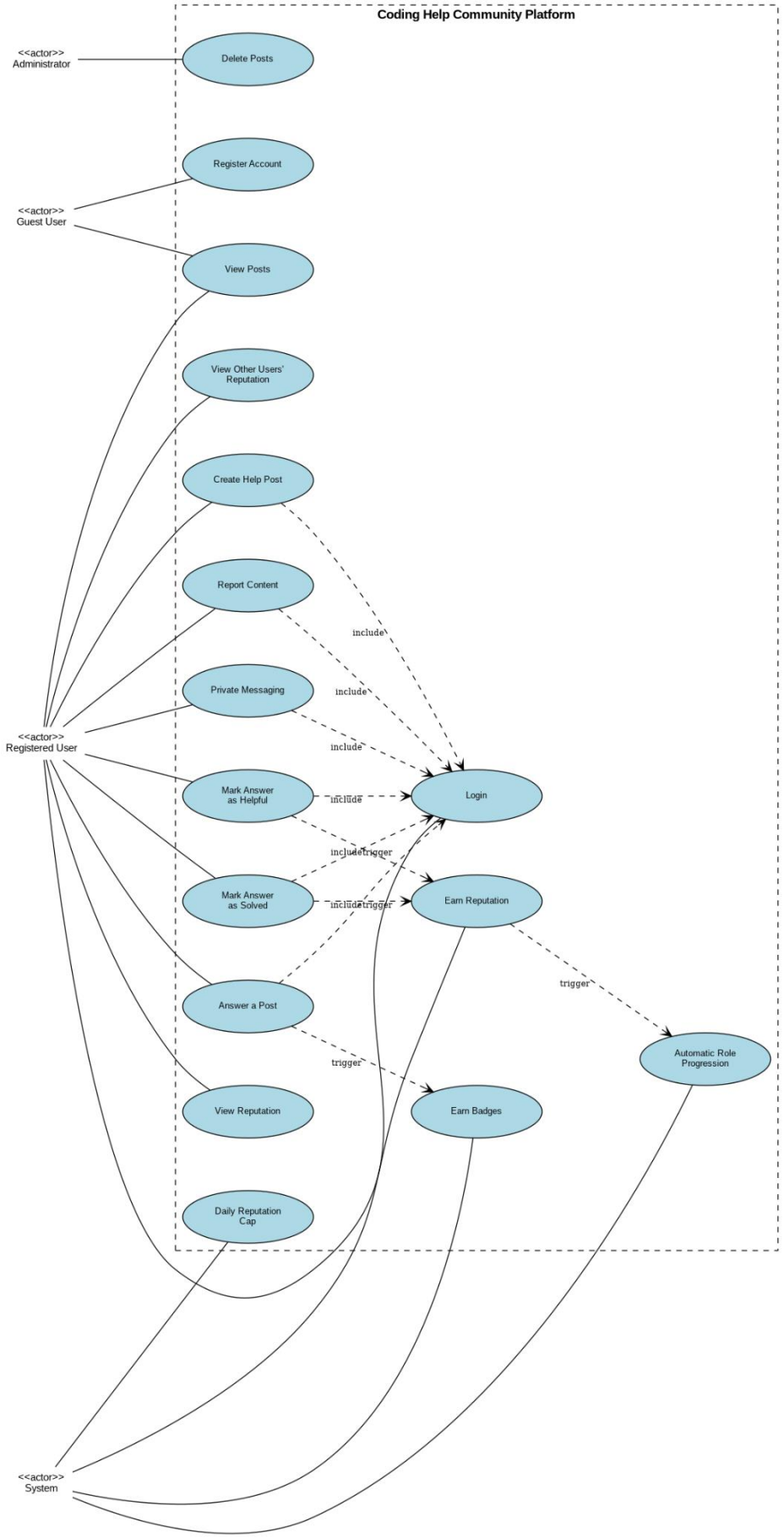
Delete Posts

As a system administrator, I want to delete inappropriate or low-quality posts, so that the community remains safe, respectful, and supportive.

Acceptance Criteria:

- Administrator must be logged in with admin privileges.
- Administrator can delete any post from the system.
- Deleted posts are removed from the public feed.
- Associated answers/comments are also removed or handled appropriately.
- The system records that the post was deleted.

Use Case Diagram



Overview

This diagram outlines the key actors, use cases, and automated system behaviours for the Coding Help Community Platform — an error handling helpline where junior developers seek help and experienced engineers provide guidance.

Actor

Actor	Key Capabilities
Guest User	Register account | View posts (read-only)
Registered User	Login | Create help post | Answer posts | Mark as solved/helpful | View reputation | Private messaging | Report content
Administrator	Delete posts | Review reported content | Configure system settings
System	Auto-calculate reputation | Role progression | Award badges | Enforce daily reputation cap

Use Cases

#	Use Case	Category
1	Register Account	Authentication
2	Login	Authentication
3	Create Help Post	Content Management
4	View Posts	Content Management
5	Answer a Post	Content Management
6	Mark Answer as Solved	Content Management
7	Mark Answer as Helpful	Content Management
8	Earn Reputation	Gamification
9	View Reputation	Gamification
10	Automatic Role Progression	Gamification
11	Earn Badges	Gamification
12	Private Messaging	Communication
13	Report Content	Moderation
14	Delete Posts	Moderation
15	Daily Reputation Cap	Moderation

Key Relationships

Include (Login required)	Create Post | Answer Post | Mark Solved | Mark Helpful | Private Messaging | Report Content
Triggers (automated)	Mark Solved -> Earn Reputation -> Role Progression | Answer Post -> Earn Badges

Reputation & Role Progression

Solved answer	+10 points
Helpful answer	+5 points
Daily cap	Max 100 points per day (resets every 24 hrs)
Contributor	0 - 99 points
Trusted Helper	100 - 499 points
Community Mentor	500+ points

User Flow Examples

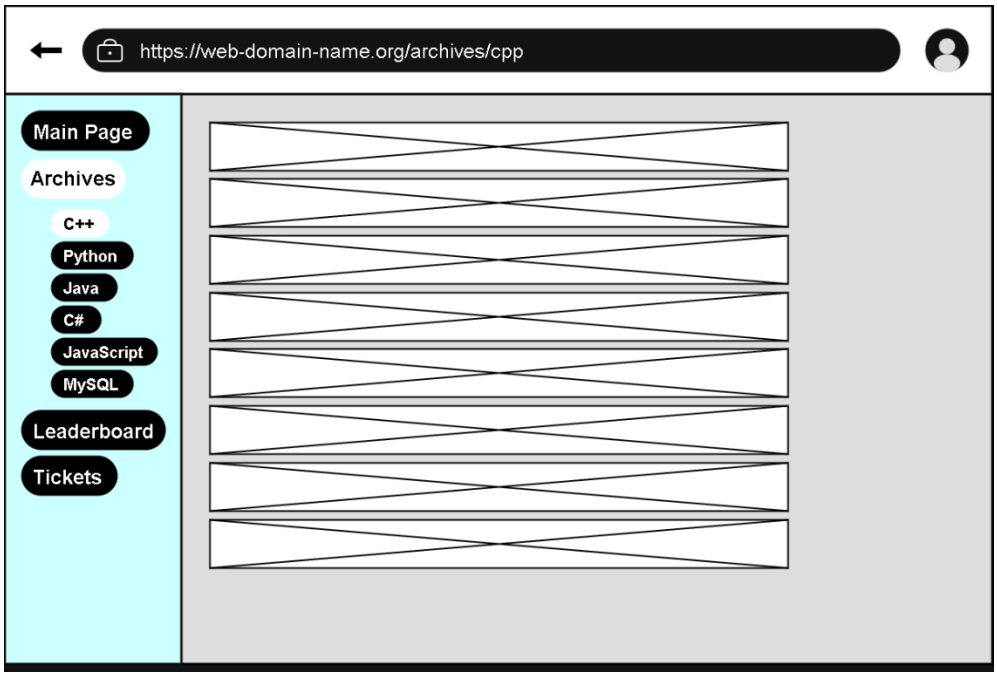
New User	Register -> Login -> View Posts -> Answer Post -> System awards Reputation & 'First Answer' badge
Posting a Problem	Login -> Create Help Post -> Community responds -> Mark Answer as Solved -> System awards Reputation & checks Role Progression
Moderation	User reports content -> Admin reviews -> Admin deletes post if violation confirmed

Implementation Notes

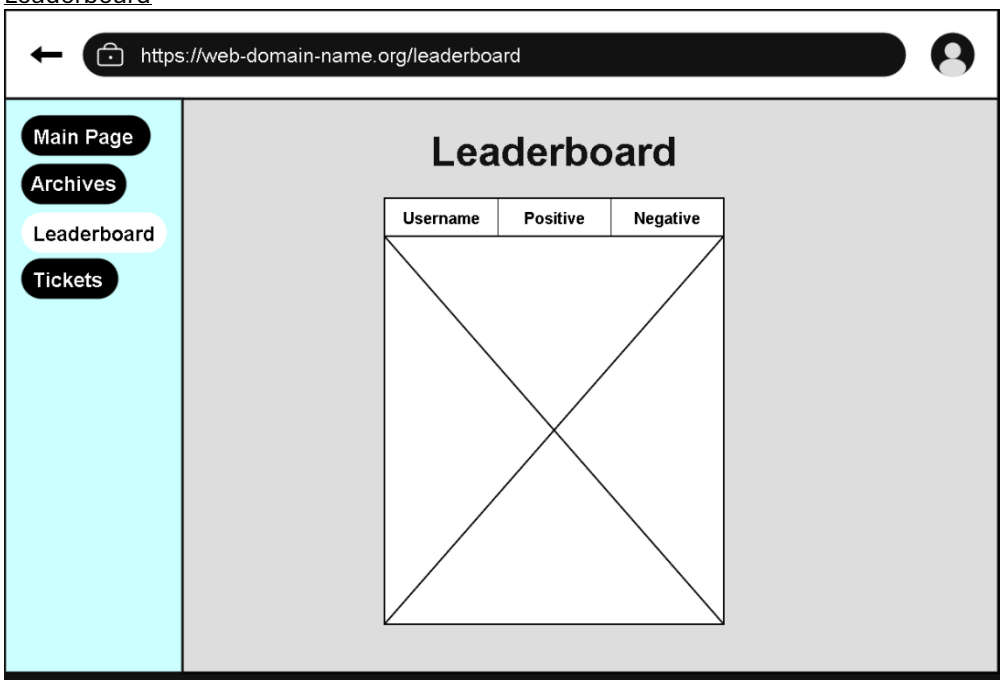
- All authenticated actions must verify user session/token.
- Reputation, badges, and role updates should run asynchronously to avoid blocking.
- Reputation calculations must be atomic to prevent race conditions.
- Rate-limit Report Content to prevent abuse.
- All admin actions should be logged for audit trail.

Wireframes

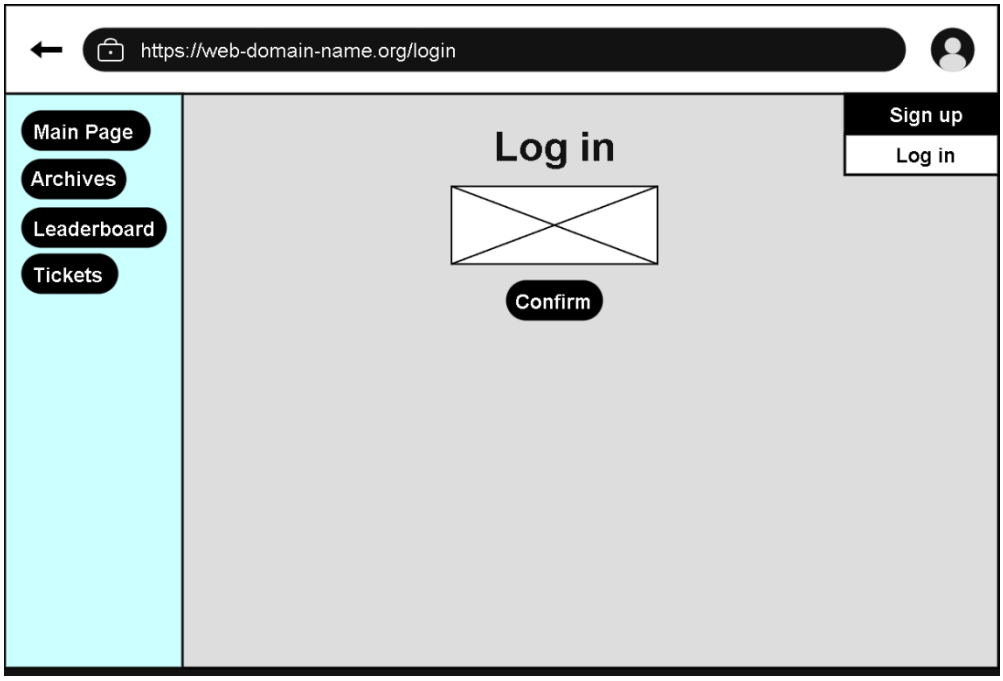
Archive Category's Tickets



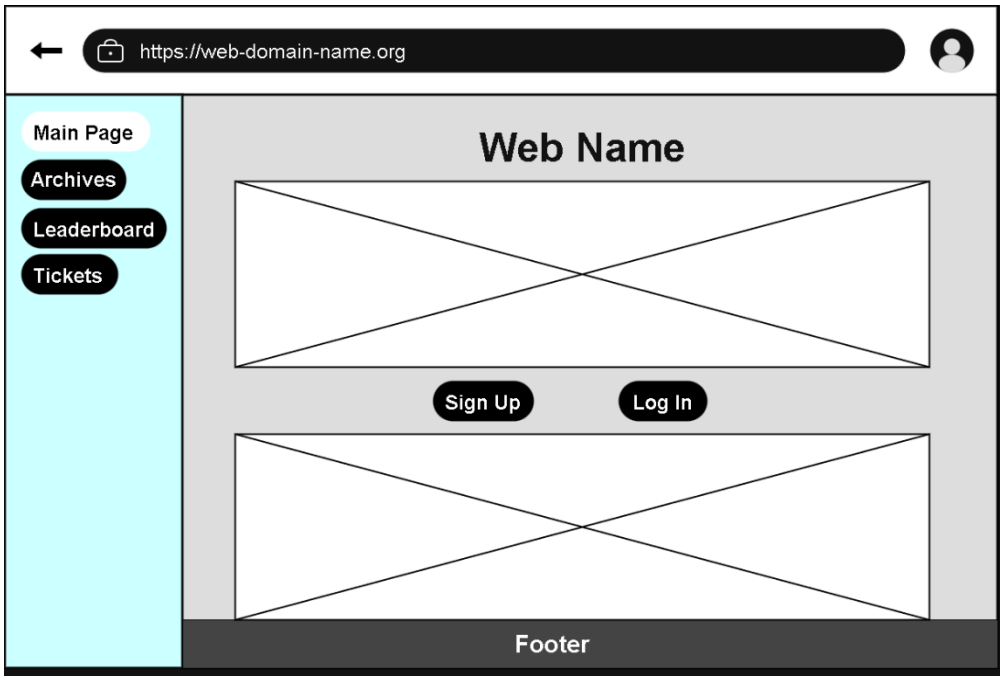
Leaderboard



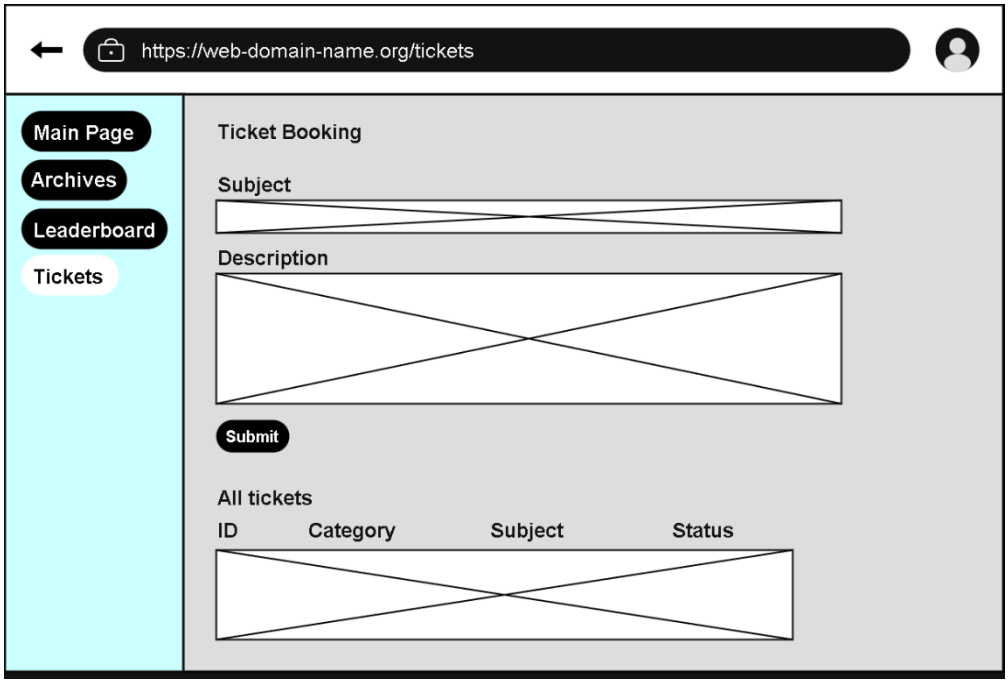
Login



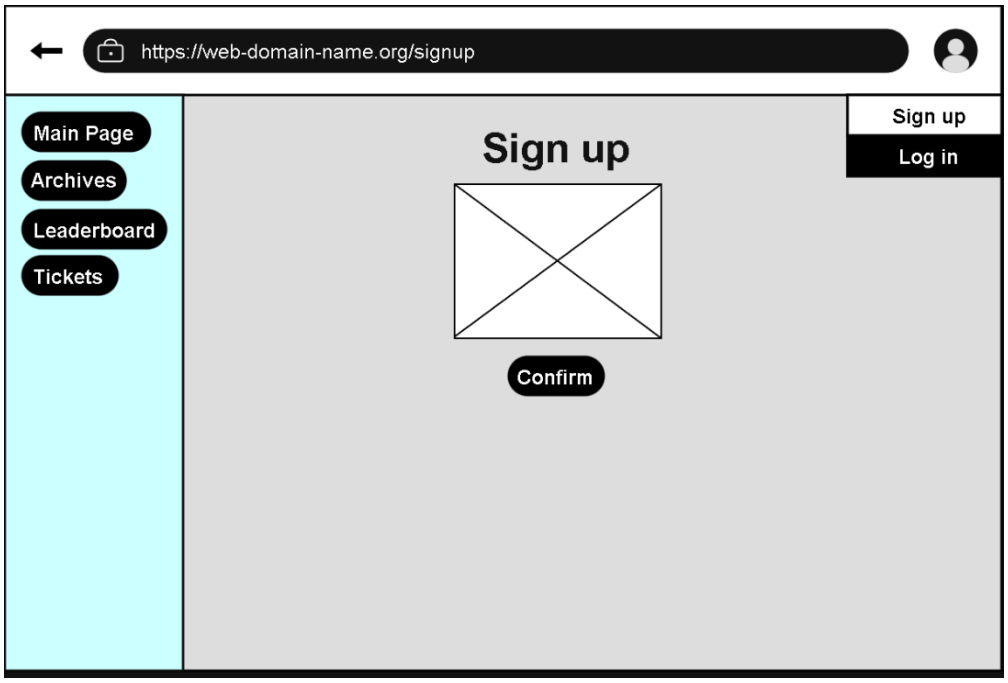
Main Page



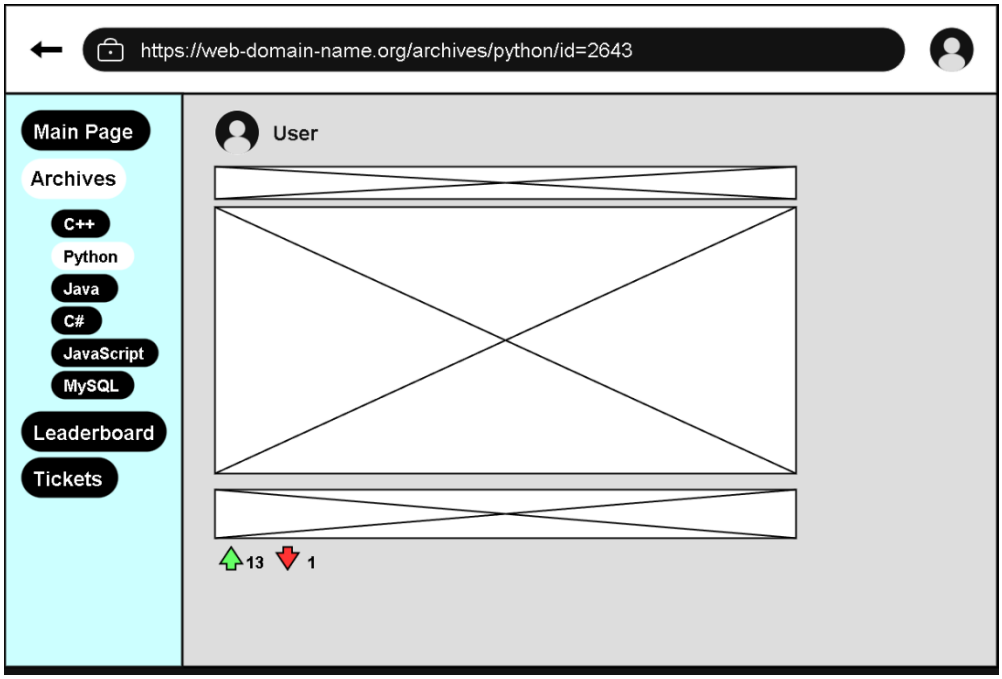
Ticket Booking



Sign up

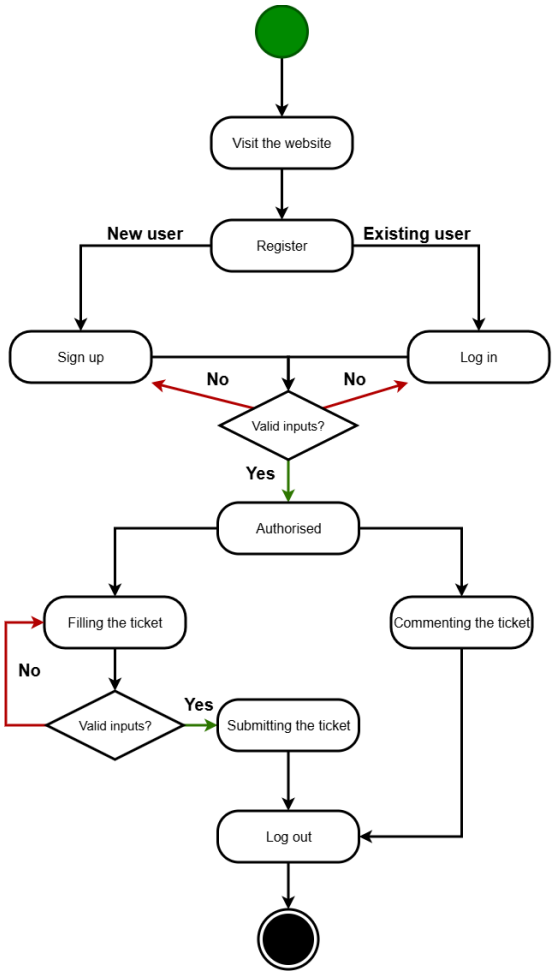


Ticket Interface

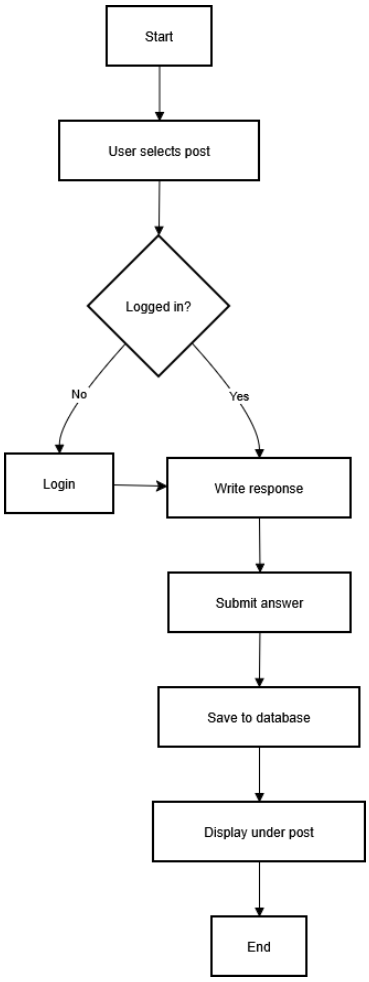


Activity Diagrams

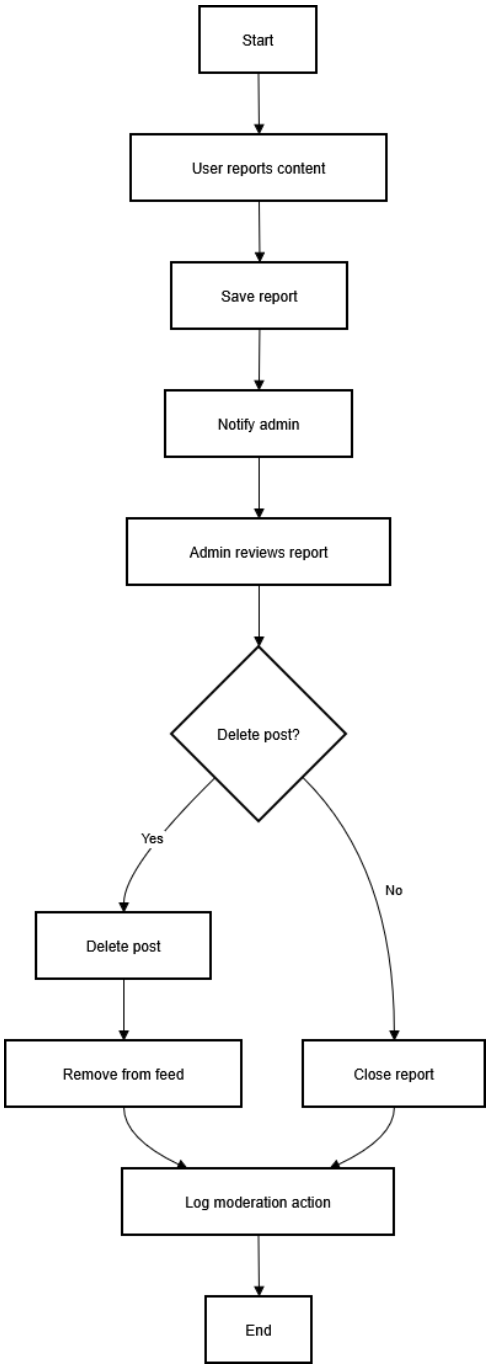
Create a Ticket (Post)



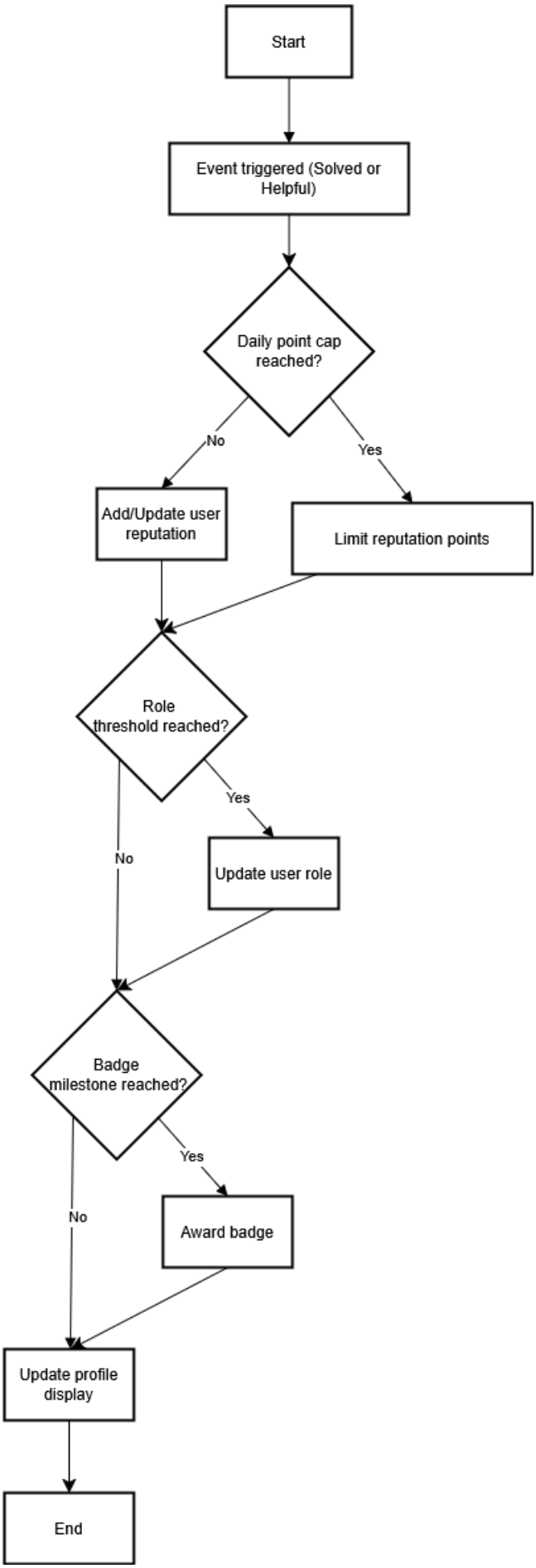
Comment on a Ticket



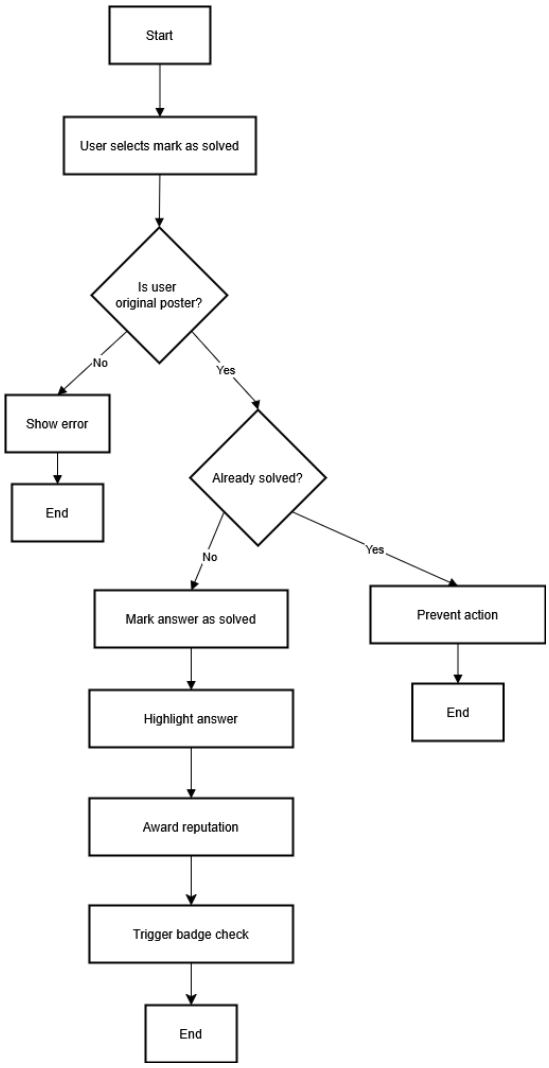
Report content



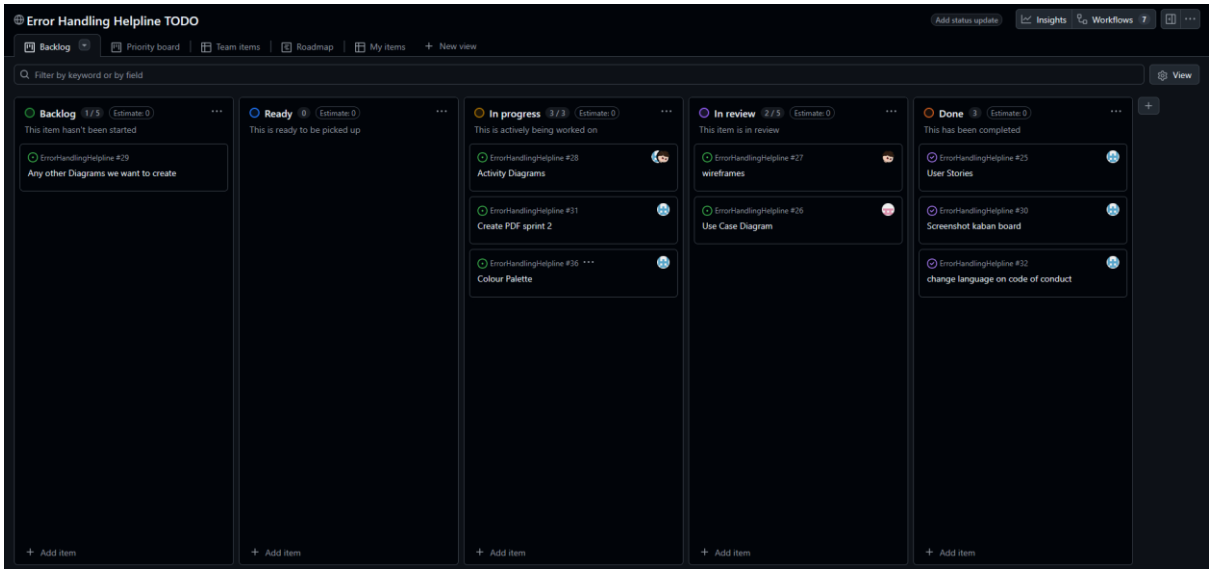
Reputation system



Mark as solved



Kanban board



Meeting Minutes

Date and Time	10/02/2026
Project Name	Error Handling Helpline
Meeting Goal	• Define some new features that relate to the coursework brief • Assign jobs for sprint 2
Facilitator	Reece Willoughby
Note taker	Reece Willoughby
Attendees	**Reece Willoughby** **Alexi Colley** **Nawaraj Budhathoki**
Roundtable Updates (each group member to contribute)	Reece suggested that we talk about the different features we should have in the project.

Discussion points	**Social messaging** **Time trials – for error handling** **Liking things (like comments)** **Give awards to smart comments.** **Give variable points to users when commenting** **Upload images / programming files** **Leaderboards – Special rewards for top 10**
Actions (list tasks and assign a group member)	Distributed all tasks.